Project Report for Deep Vision

“ Segmentation with CNNs: UNet vs. SegNet. Find the differences and limitations.”

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Abstract

This report explores the field of semantic image segmentation using convolutional neural networks (CNNs). We specifically investigate and compare two popular deep learning architectures: UNet and SegNet. The main goal was to understand their differences, strengths, and limitations when applied to real-world datasets.

Our study involved conducting various experiments on two distinct datasets: ISIC 2018 for medical image segmentation (skin lesions) and Cityscapes for urban scene understanding. For both UNet and SegNet, we experimented with different encoder models (like ResNet, EfficientNet, and VGG) and various loss functions (Cross-Entropy, Dice, and a combined approach).

The findings show that both architectures can perform image segmentation, but UNet, especially when using pre-trained encoders, generally achieved higher accuracy and better detail, particularly on the complex Cityscapes dataset. SegNet, when trained from scratch, showed limitations in capturing fine details and achieving high performance within limited training epochs. This report highlights the practical challenges and trade-offs involved in choosing and training deep learning models for image segmentation tasks.

1 Introduction

In the modern digital age, images are an essential part of our daily lives, and the ability to understand their content at a detailed level is becoming increasingly important for many advanced technologies. One fundamental area in computer vision that enables this deep understanding is image segmentation. This process involves precisely dividing a digital image into different meaningful parts or regions, where each individual pixel is assigned to a specific object or area of interest. For example, in an autonomous driving scenario, image segmentation can accurately delineate every car, pedestrian, road, and building, providing a crucial pixel-by-pixel understanding of the environment. This level of detail goes significantly beyond simpler tasks like image classification, which only identifies the presence of an object.

The field of image processing experienced a major paradigm shift with the emergence of Convolutional Neural Networks (CNNs) [1]. Their inherent ability to automatically learn complex and hierarchical visual features directly from raw image data, without the need for manual feature engineering, has made them exceptionally powerful for various computer vision tasks, including the intricate demands of segmentation. Among the numerous successful CNN architectures developed for this purpose, UNet[2] and SegNet[3] stand out as two highly influential and widely adopted models specifically designed for semantic segmentation. Semantic segmentation aims to classify every single pixel in an image into one of several predefined categories (e.g., "vehicle," "road," "sky," "person"). This project focuses on a direct exploration and comparative analysis of these two prominent architectures, given their distinct design principles and widespread application in diverse fields.

2 Motivation

This project's core motivation was to thoroughly compare the UNet and SegNet architectures, understanding how their unique designs impact their practical performance in image segmentation. We aimed to identify their strengths and limitations by conducting experiments that varied key components, such as the choice of encoder and the loss function used during training. To ensure a comprehensive study, we applied these models to two diverse and challenging datasets: the ISIC 2018 dataset for medical image segmentation (skin lesions), where precision is critical for diagnosis, and the Cityscapes dataset for complex urban scene segmentation, which is essential for developing self-driving car technologies. Through these practical comparisons, this report seeks to provide insights into building effective deep learning models for varied segmentation applications.

3 Related Work

Image segmentation has long been a key challenge in computer vision[1]. Early methods relied on traditional algorithms like edge detection, but they struggled with complex images[4]. A major step forward came with Convolutional Neural Networks (CNNs)[5]. In 2015, Fully Convolutional Networks (FCNs) revolutionized the field by enabling pixel-level prediction directly from images, paving the way for more advanced architectures.

Following FCNs, two important encoder-decoder architectures emerged to improve segmentation quality, especially by handling the loss of detail during downsampling. **SegNet**, introduced in 2017, innovated by using "pooling indices" to precisely reconstruct feature maps during decoding, helping to recover fine-grained details[6]. Around the same time, **UNet**, developed in 2015 for medical imaging, became known for its "skip connections." These connections combine high-level semantic information from deep layers with fine spatial details from earlier layers, leading to very precise segmentations[6].

The availability of large datasets, such as Cityscapes and ISIC 2018, has also been crucial. Cityscapes offers complex urban scenes, pushing models to understand diverse objects in real-world conditions. ISIC 2018 focuses on critical medical applications, demanding high accuracy for diagnostic purposes. These datasets allow for thorough training and testing of segmentation models[7].

This project builds on this work by directly comparing SegNet and UNet on these challenging datasets. We experiment with different encoder backbones and loss functions to better understand their practical performance and limitations in various segmentation scenarios.

4 Software and Tools

The core development for this image segmentation project was carried out in Python, utilizing its extensive machine learning ecosystem. For building, training, and evaluating our deep learning models, we primarily relied on PyTorch, a powerful open-source framework known for its flexibility. Data handling and fundamental image processing tasks were managed using OpenCV (cv2)[8] and NumPy, which provided efficient operations for image and mask files. All code development, experimentation, and analysis were performed within Jupyter Notebooks, offering an interactive and organized environment.

To enhance our models' robustness, Albumentations was extensively used for various image augmentation techniques, such as resizing, flipping, and color adjustments. For the UNet architecture, the `segmentation_models_pytorch` (smp)[9] library proved invaluable, providing pre-built models with pre-trained encoder backbones to accelerate experimentation. Matplotlib was employed for visualizing training progress and displaying segmentation outputs. Finally, to meet the demanding computational requirements of deep learning, we leveraged

CUDA, enabling the use of a Graphics Processing Unit (GPU) for significantly faster model training.

5 Methodology

This section describes the practical steps and setup used to conduct the image segmentation experiments. It covers the datasets, how the data was prepared, the details of the neural network models, the training strategies, and how the models were evaluated.

5.1 Datasets

To properly test the performance and limitations of the UNet and SegNet architectures, we chose two very different datasets. Using diverse datasets helps us understand how well the models work in various real-world situations, from medical images to complex outdoor scenes.

5.1.1 ISIC 2018: Skin Lesion Analysis Towards Melanoma Detection

Characteristics: The ISIC 2018 dataset[10] is a collection of dermoscopic images of skin lesions. It is specifically designed for medical image analysis, with a focus on helping to detect melanoma, a type of skin cancer. The images are typically high-resolution, and each image comes with a corresponding binary mask. This mask clearly separates the lesion (the area of interest) from the surrounding healthy skin.

Purpose in Project: We chose this dataset to evaluate how well UNet and SegNet perform in a critical medical application. In medical imaging, precise segmentation is very important for accurate diagnosis and treatment planning. This dataset allowed us to test the models' ability to identify subtle boundaries and variations in a relatively uniform background. The segmentation task here is simpler in terms of the number of classes (usually just two: lesion and non-lesion), but it demands high accuracy due to its medical importance.

5.1.2 Cityscapes Dataset

Characteristics: The Cityscapes dataset[11] is much larger and more complex. It consists of a diverse collection of images taken from street scenes in various cities. Unlike ISIC, Cityscapes provides fine-grained pixel-level annotations for a wide range of urban elements, such as roads, sidewalks, buildings, vehicles (cars, trucks, buses), pedestrians, traffic signs, and vegetation. It includes images captured in different weather conditions and times of day, making it a challenging dataset for segmentation. The images are typically high-resolution, requiring models to process a lot of visual information. The dataset has 19 different semantic classes, plus an "ignore" label for pixels that are not part of any defined class.

Group	Classes
flat	road · sidewalk · parking^{+} · rail track^{+}
human	person* · rider*
vehicle	car* · truck* · bus* · on rails* · motorcycle* · bicycle* · caravan^{*+} · trailer^{*+}
construction	building · wall · fence · guard rail^{+} · bridge^{+} · tunnel^{+}
object	pole · pole group^{+} · traffic sign · traffic light
nature	vegetation · terrain
sky	sky
void	ground^{+} · dynamic^{+} · static^{+}

Figure 1: Class Definitions
[11]

Purpose in Project: This dataset was chosen to evaluate the models' ability to handle complex, multi-class segmentation tasks in a real-world environment. It pushes the models to distinguish between many different objects and to accurately segment them even when they are close together or partially hidden. Its complexity allowed us to thoroughly compare UNet and SegNet in a scenario typical for applications like autonomous driving and urban planning, where detailed scene understanding is vital.

5.2 Data Preprocessing and Augmentation

Before feeding the images into our neural networks, it was crucial to perform several preprocessing and augmentation steps. These steps are vital for two main reasons: first, to standardize the input data for the models, and second, to increase the diversity of the training data, which helps the models learn more robust features and prevents them from simply memorizing the training examples (overfitting). We primarily used the Albumentations library for these transformations, known for its efficiency and wide range of augmentation options.

5.2.1 Core Preprocessing Steps

For both the ISIC 2018 and Cityscapes datasets, the following fundamental steps were applied to all images and their corresponding masks:

Resizing: All input images and their masks were resized to a consistent dimension. For the Cityscapes dataset, images were resized to 384×768 pixels. For the

ISIC 2018 dataset, images were resized to 256×256 pixels. This standardization ensures that all inputs to the network have the same shape, which is necessary for batch processing.

Normalization: Image pixel values were normalized using the mean and standard deviation derived from the ImageNet dataset. This process scales pixel intensities to a standard range, which helps in stabilizing and accelerating the training process of deep learning models. The mean values used were (0.485,0.456,0.406) and standard deviations were (0.229,0.224,0.225) across the RGB channels.

Tensor Conversion: After resizing and normalization, the images and masks were converted into PyTorch tensors. Images were transformed into a channel-first format (Channels, Height, Width), while masks were converted to Long type tensors, suitable for categorical cross-entropy loss functions.

Mask Remapping (Cityscapes Specific): For the Cityscapes dataset, the original mask labels (0-33 with various ignore IDs) were remapped to a continuous range of 0 to 18 for the 19 target classes, with an ignore index of 255 for unlabeled pixels. This ensures that only relevant classes contribute to the loss calculation.

5.2.2 Data Augmentation Techniques

To enhance model generalization and prevent overfitting, a variety of augmentation techniques were applied only to the training datasets. These techniques simulate real-world variations and expose the model to a wider range of scenarios.

For Geometric Transformations, images and masks were randomly flipped horizontally (50 persantage chance) to introduce lateral variations. Further, random shifts, scaling, and rotations were applied using ShiftScaleRotate to make the model robust to changes in object position, size, and orientation. Non-linear deformations were also introduced through Optical Distortion and Grid Distortion, simulating effects like lens distortions or flexible object shapes, thereby increasing data diversity.

Regarding Color and Brightness Adjustments, ColorJitter and RandomBrightnessContrast were used to apply random changes in brightness, contrast, saturation, and hue, making the model resilient to varying lighting conditions. Additionally, GaussNoise added random Gaussian noise to images to simulate sensor noise, and CLAHE (Contrast Limited Adaptive Histogram Equalization) was applied to enhance local contrast, making features more discernible.

5.3 Model Architectures

At the heart of our image segmentation project are two prominent deep learning architectures: UNet and SegNet. Both models follow an encoder-decoder structure, which is a common design for segmentation tasks. In this structure, the "encoder" part processes the input image to extract important features and

reduce its spatial size, while the "decoder" part then uses these features to reconstruct the segmentation mask at the original image resolution. The main differences between UNet and SegNet lie in how they transfer information from the encoder to the decoder.

5.3.1 UNet

For this project, the UNet architecture was chosen as one of our primary models for image segmentation. Its well-known U-shaped design, consisting of a symmetric contracting path (encoder) and an expanding path (decoder), was central to our approach[12].

In the Encoder (Contracting Path), we used standard convolutional layers and pooling operations to extract features and reduce the spatial dimensions of the input images. This process helped our network learn high-level, abstract features. For our experiments, we specifically utilized pre-trained encoders available from the segmentation models pytorch library. These included ResNet34, known for its robust performance; VGG16, a classic and straightforward choice; and EfficientNetB0, selected for its balance of accuracy and computational efficiency. By using these encoders, which were pre-trained on a large dataset like ImageNet, we applied transfer learning. This allowed our UNet models to start with a strong understanding of general image features, significantly speeding up our training process and often leading to better performance, especially given the specific nature of our datasets[5].

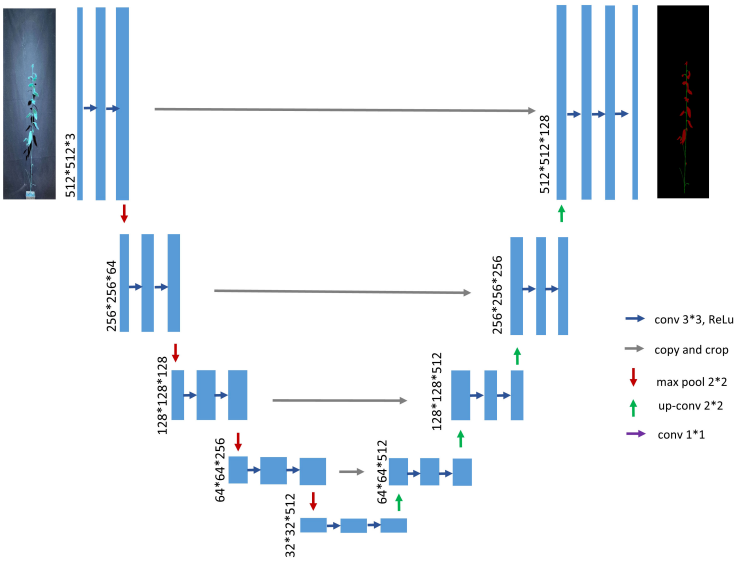

Figure 2: CNN-based unet architecture

The Decoder (Expanding Path) in our UNet implementation was responsible for reconstructing the segmentation map from the features learned by the encoder. This was achieved using upsampling operations, which increased the spatial dimensions of the feature maps. A critical aspect of our UNet models was the use of skip connections. These connections directly linked feature maps from the encoder path to the decoder path at the same resolution level.

This design choice was crucial for our project because it enabled the decoder to combine two vital types of information: the high-level semantic understanding from the deeper encoder layers (telling what an object is) and the fine-grained spatial details from the shallower encoder layers (telling where an object is and its precise boundaries). This combination allowed our UNet models to produce very precise segmentation masks with accurate boundaries, which was essential for both the medical (ISIC 2018) and urban scene (Cityscapes) segmentation tasks [5].

5.3.2 SegNet

The SegNet architecture was the second primary model we employed in this project. While it also uses an encoder-decoder structure, its approach to transferring information from the encoder to the decoder differs significantly from UNet's[12].

In our project's Encoder for SegNet, similar to UNet, we extracted features through a series of convolutional layers and max-pooling operations, progressively reducing the spatial resolution of the input images. A key decision for our SegNet implementation was to use a custom VGG-like encoder. This means that while its structural design was inspired by the well-known VGG network, this encoder was specifically built and trained from scratch as part of our experiments. We deliberately chose not to use pre-trained weights for this encoder. This approach allowed us to directly observe SegNet's inherent performance without the benefits of transfer learning, providing a clear comparison to how UNet typically utilizes pre-trained encoders in our study[2][13].

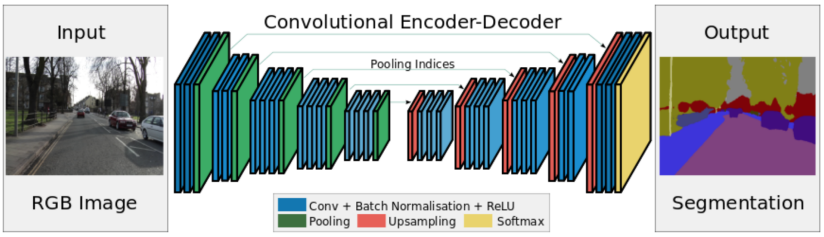

Figure 3: Cnn based Segnet architecture

The Decoder in our SegNet model was designed to upsample the feature maps back to the original image resolution, thereby reconstructing the detailed seg-

mentation mask. The crucial mechanism for this reconstruction in our SegNet implementation was the use of pooling indices.

During the max-pooling operation in the encoder phase of our SegNet, the exact locations (indices) of the maximum values within each pooling window were recorded. These "pooling indices" were then passed forward to the corresponding upsampling layer in the decoder. When the decoder performed upsampling, our SegNet model utilized these stored indices to place the values from the lower-resolution feature map back into their original, precise positions. This effectively performed a non-linear unpooling operation. For our project, this method was important as it helped in preserving fine boundary information in the segmentation masks and contributed to SegNet's computational efficiency by reducing the need for extensive learnable upsampling layers.

5.3.3 4.4 Loss Functions

- **Cross-Entropy Loss:** This is a very common loss function for pixel-wise classification. For a given pixel, if N is the number of classes, y_c is the true binary indicator (1 if class c is the true class, 0 otherwise), and p_c is the predicted probability for class c, the weighted Cross-Entropy Loss (L_{CE}) is calculated as:

$$L_{CE} = -\frac{1}{M}\sum_{i=1}^{M}\sum_{c=1}^{N} w_c \cdot y_{i,c} \log(p_{i,c})$$

  where M is the number of pixels, and w_c are the class weights used to address class imbalance.

- **Dice Loss:** Unlike Cross-Entropy, Dice Loss directly measures the overlap between the predicted segmentation mask and the ground truth mask. It is derived from the Dice coefficient. For a given class c, let P_c be the set of predicted pixels for class c and G_c be the set of ground truth pixels for class c. The Dice coefficient (DSC) for that class is:

$$DSC_c = \frac{2|P_c \cap G_c|}{|P_c| + |G_c|}$$

  The Dice Loss (L_{Dice}) is typically defined as $1 - DSC$, averaged over all classes (or a multiclass variant). We applied a softmax activation to the model's output before calculating the Dice Loss, as Dice Loss typically expects probability-like inputs.

- **Combined Cross-Entropy and Dice Loss:** To leverage the strengths of both loss functions, we also used a combined approach, which is a weighted sum of the Cross-Entropy Loss and the Dice Loss. The combined loss ($L_{Combined}$) is given by:

$$L_{Combined} = \alpha \cdot L_{CE} + \beta \cdot L_{Dice}$$

where α and β are weighting factors. For some experiments, we used a tuned version of this combined loss, specifically setting $\alpha = 0.3$ and $\beta = 0.7$ to emphasize the Dice Loss's benefits for boundary accuracy and class imbalance.

5.4 Training Process

All models were trained using the Adam optimizer with an initial learning rate of 0.0001 and a weight decay of 1×10 4 for regularization. A ReduceLROnPlateau scheduler was used, which reduced the learning rate if the validation mIoU did not improve for 3 epochs. Each model was trained for 80 epochs with a batch size of 8. Training was accelerated using a CUDA-enabled GPU.

5.5 Evaluation Metrics

- **Pixel Accuracy:** This metric calculates the percentage of correctly classified pixels. While simple, it can be misleading with class imbalance.
- **Mean Intersection over Union (mIoU):** This is the primary metric, calculated as the average IoU score across all classes. IoU for a single class c measures the overlap between the predicted mask (P_c) and the ground truth mask (G_c), defined as:

$$IoU_c = \frac{|P_c \cap G_c|}{|P_c \cup G_c|}$$

  The mIoU is then the average of the IoU scores for all N relevant classes:

$$mIoU = \frac{1}{N} \sum_{c=1}^{N} IoU_c$$

  It is a more robust indicator of segmentation quality, especially for datasets with varying object sizes and class distributions, as it directly reflects boundary accuracy.

6 Experiments and Results

This section presents the outcomes of our image segmentation experiments, detailing the performance of the UNet and SegNet architectures across the ISIC 2018 and Cityscapes datasets. We evaluated various encoder backbones and loss functions to identify optimal configurations.

6.1 Experimental Setup Overview

Our experiments involved training UNet and SegNet models on two distinct datasets: ISIC 2018 for medical image segmentation (binary classification) and Cityscapes for complex urban scene segmentation (multi-class classification).

Models were trained using the Adam optimizer with a learning rate scheduler, and performance was primarily assessed using Mean Intersection over Union (mIoU) and Pixel Accuracy.

6.2 Results on ISIC 2018 Dataset

The ISIC 2018 dataset experiments aimed to evaluate segmentation performance on medical images. Tables 1 and 2 summarize the key metrics for UNet and SegNet models, respectively.

Encoder	**Loss**	**Best Val IoU**	**Test Pix Acc**	**Test mIoU**
VGG16	CE	0.846	0.932	0.840
VGG16	Dice	0.767	0.885	0.745
VGG16	Combined	**0.850**	**0.934**	**0.847**
ResNet34	CE	**0.852**	**0.934**	0.843
ResNet34	Dice	0.793	0.901	0.777
ResNet34	Combined	0.850	0.933	0.842

Table 1: UNet Performance on ISIC 2018 Dataset

Analysis: On the ISIC dataset, UNet models generally outperformed SegNet models across all configurations, achieving higher mIoU and pixel accuracy. The UNet with ResNet34 encoder and Cross-Entropy loss achieved the highest validation mIoU (0.852), closely followed by UNet with VGG16 and combined CE+Dice loss (0.850). For both architectures, the combined Cross-Entropy and Dice loss often yielded competitive or superior results compared to individual loss functions, demonstrating the benefit of combining their strengths. UNet models also showed faster inference times compared to SegNet models with ResNet34 encoders.

Encoder	**Loss**	**Best Val IoU**
VGG16	CE	0.793
VGG16	Dice	0.779
VGG16	Combined	**0.804**
ResNet34	CE	0.795
ResNet34	Dice	0.745
ResNet34	Combined	0.794
EfficientNet-B3	CE	0.789
EfficientNet-B3	Dice	0.702
EfficientNet-B3	Combined	0.789

Table 2: SegNet Performance on ISIC 2018 Dataset (Best Validation IoU)

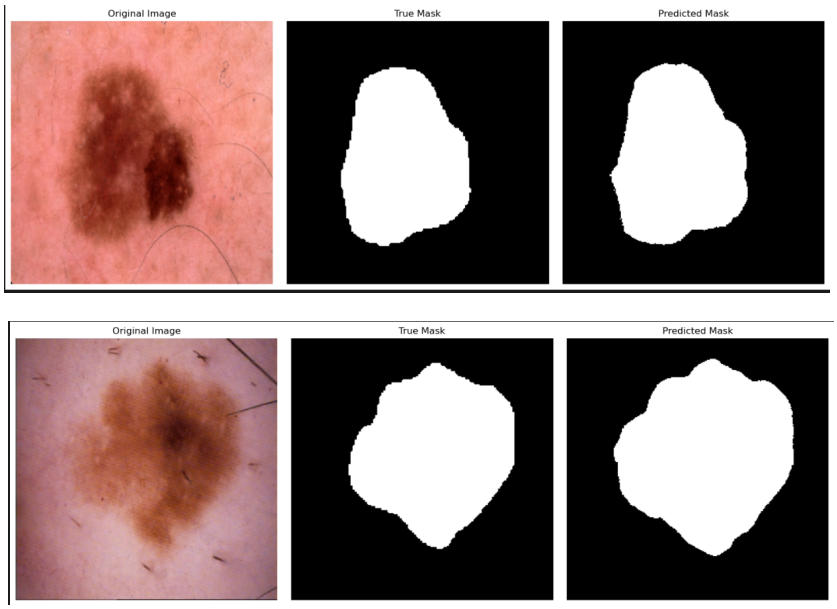

Figure 4: Segmentation of ISIC dataset

6.3 Results on Cityscapes Dataset

Experiments on the Cityscapes dataset presented a more complex multi-class segmentation challenge. Table 3 summarizes the performance of UNet models, while Table 1 highlights the SegNet results.

Encoder	**Loss**	**Best Val IoU**	**Train Time (s)**
ResNet34	CrossEntropy	0.407	9827.5
EfficientNet-B3	Dice	0.340	11324.6
VGG16	Combined	**0.439**	13600.1

Table 3: UNet Performance on Cityscapes Dataset (Validation IoU)

Encoder	**Loss**	**Val mIoU**	**Val Pix Acc**
VGG16_Custom	Combined CE+Dice	**0.421**	**0.910**
VGG16_Custom	CrossEntropy	0.259	0.871
VGG16_Custom	Dice	0.277	0.878

Table 4: SegNet Performance on Cityscapes Dataset (Validation Metrics)

Analysis: On Cityscapes, the UNet with VGG16 encoder and combined CE+Dice loss achieved the highest validation mIoU among the tested UNet configurations (0.439). The SegNet model, particularly with the custom VGG16 encoder and

tuned combined CE+Dice loss, also showed competitive performance (0.421 mIoU). It's notable that the Cityscapes dataset, being more complex with 19 classes and diverse scenes, resulted in generally lower mIoU scores for both architectures compared to the binary ISIC task. The training times for Cityscapes models were significantly longer due to the larger image dimensions and dataset size.

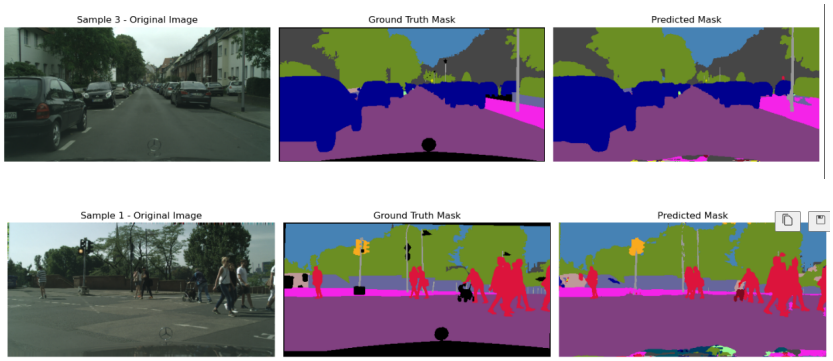

Figure 5: Segmentation of ISIC dataset

6.4 Discussion of Findings

Our experiments reveal several key insights into UNet and SegNet performance on different segmentation tasks:

- **UNet's Superiority:** UNet consistently demonstrated stronger performance, particularly on the ISIC dataset, often achieving higher mIoU and pixel accuracy. This is largely attributable to its effective skip connections, which preserve fine-grained spatial details, and the benefit of using pre-trained encoders (transfer learning) for initial feature extraction.
- **Impact of Loss Functions:** The combined Cross-Entropy and Dice loss proved to be a robust choice across both datasets and architectures. It often led to better results than individual loss functions, suggesting that combining pixel-wise classification with overlap optimization is beneficial, especially for handling class imbalance and boundary precision.
- **Dataset Complexity:** The performance drop from ISIC to Cityscapes highlights the increased complexity of multi-class semantic segmentation in diverse real-world scenes. Models require more robust feature learning and larger training times to generalize effectively on such datasets.
- **SegNet's Performance:** While SegNet did not consistently outperform UNet, its ability to achieve competitive results on Cityscapes (mIoU of 0.421) even with a custom encoder trained from scratch is noteworthy. This demonstrates its inherent capability for segmentation, although it

may require more extensive training or pre-trained encoders to match UNet's top performance.

- **Inference Time:** UNet models generally exhibited faster inference times, which is crucial for real-time applications like autonomous driving.

7 Conclusion

This report evaluated UNet and SegNet architectures for semantic image segmentation on the ISIC 2018 (medical) and Cityscapes (urban scenes) datasets. On the ISIC 2018 dataset, UNet consistently outperformed SegNet, achieving higher mIoU and pixel accuracy, largely due to its effective skip connections and the benefit of pre-trained encoders. For the more complex Cityscapes dataset, both architectures faced greater challenges. UNet, particularly with a VGG16 encoder and combined loss, showed strong performance. Notably, the custom SegNet, trained from scratch, also demonstrated competitive capabilities, especially with a tuned combined loss.

The combined Cross-Entropy and Dice Loss proved highly effective across all experiments, balancing pixel-wise accuracy with improved boundary delineation and handling of class imbalance. In summary, UNet remains a highly effective architecture for diverse segmentation tasks, often benefiting from transfer learning. SegNet offers a capable alternative. This study underscores the importance of selecting appropriate architectures, tuning loss functions, and employing robust data augmentation for optimal segmentation performance.

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